

HARTSVILLE, SC, USA

WMO: 720633

Lat: 34.400N

Lon: 80.117W

Elev: 111

StdP: 100.00

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00219

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.0	-2.5	-16.3	0.9	-1.2	-12.8	1.3	1.5	7.3	12.3	6.5	11.5	1.4	10	0.357

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	35.2	23.0	33.9	22.9	32.7	22.9	26.3	29.4	25.7	29.5	25.1	29.5	2.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.9	21.5	27.3	25.0	20.3	27.0	23.9	19.0	26.4	82.2	29.7	79.4	30.5	77.4	27.8	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.6	5.5	4.8	-8.2	37.2	2.7	1.7	-10.2	38.5	-11.8	39.5	-13.3	40.4	-15.3	41.7		
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	7.0	9.2	12.2	17.5	22.0	25.8	27.0	26.3	24.2	17.8	11.8	9.1
	DBStd	8.20	5.52	5.36	5.27	4.16	3.27	2.26	1.67	1.83	2.71	4.29	4.20	5.16
	HDD10.0	347	125	74	37	3	0	0	0	0	0	1	27	80
	HDD18.3	1433	352	258	199	64	10	0	0	0	1	62	198	288
	CDD10.0	3103	32	52	107	227	373	474	528	507	424	243	82	53
	CDD18.3	1149	1	3	11	39	125	224	270	248	176	45	3	3
	CDH23.3	10419	7	21	108	406	1092	2164	2614	2198	1442	341	19	6
	CDH26.7	4493	0	1	17	106	411	1023	1256	997	600	83	1	0
Wind	WSAvg	1.8	1.8	2.1	2.1	2.2	1.8	1.5	1.5	1.4	1.8	1.8	1.8	1.7
Precipitation	PrecAvg	1145	86	86	90	77	80	123	134	134	119	79	72	83
	PrecMax	1350	192	175	182	143	145	226	212	232	256	148	164	201
	PrecMin	769	25	32	11	23	36	50	50	48	16	0	12	21
	PrecStd	143	49	41	41	35	27	44	52	60	67	40	38	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.1	26.1	28.6	31.2	34.7	37.4	37.2	36.0	34.9	32.0	26.1	23.9
		MCWB	17.2	17.9	17.2	18.4	21.5	23.4	23.5	22.0	21.1	22.9	18.9	19.5
	2%	DB	20.9	22.8	26.3	29.8	32.4	35.1	35.2	34.1	32.9	28.7	22.8	22.1
		MCWB	15.5	16.3	16.6	19.0	20.8	23.4	23.5	22.9	22.2	21.7	16.6	18.7
	5%	DB	17.8	21.0	23.9	27.6	31.1	32.9	33.8	32.7	32.2	27.1	21.3	20.0
		MCWB	14.1	15.3	15.6	18.1	20.8	23.0	23.5	23.1	21.9	20.1	16.4	17.3
	10%	DB	16.0	17.8	22.0	26.0	28.9	32.2	32.5	32.1	30.9	25.1	19.8	17.6
		MCWB	12.2	13.5	14.8	17.3	20.7	22.7	23.6	23.2	22.1	19.2	15.5	15.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.1	19.2	19.5	22.7	24.4	26.4	27.4	27.3	26.0	24.9	20.8	21.0
		MCDB	22.0	22.9	23.8	25.7	27.7	29.9	30.6	30.0	28.9	27.8	21.8	22.2
	2%	WB	17.5	18.0	18.4	21.6	23.1	25.5	26.4	26.2	24.8	23.8	19.3	19.5
		MCDB	19.2	21.2	23.6	25.2	27.2	30.1	29.4	29.4	29.1	27.2	21.7	21.3
	5%	WB	15.2	16.8	17.5	20.3	22.6	24.8	25.7	25.5	23.9	22.7	17.9	17.6
		MCDB	16.9	19.9	21.0	24.2	26.9	29.9	29.7	29.2	28.4	25.2	19.7	18.3
	10%	WB	12.9	14.8	16.4	19.0	22.0	24.1	25.2	25.0	23.2	21.2	16.5	15.8
		MCDB	14.6	17.7	20.2	24.1	26.6	29.4	29.8	29.1	27.5	23.1	18.5	17.4
Mean Daily Temperature Range	5% DB	MDBR	10.9	11.5	12.6	13.5	12.2	12.1	11.6	10.8	11.3	11.8	12.0	10.2
		MCDBR	13.6	13.5	15.3	15.8	14.9	13.9	13.4	12.5	13.6	13.3	13.5	11.1
		MCWBR	8.3	8.2	7.1	6.5	5.1	4.4	4.5	4.4	4.6	6.0	8.1	7.3
	5% WB	MCDBR	11.0	12.4	12.7	11.8	11.5	12.3	11.5	10.5	10.9	10.0	10.8	9.4
		MCWBR	8.9	8.9	6.9	6.5	4.8	4.7	4.5	4.1	4.4	5.7	7.7	8.0
Clear-Sky Solar Irradiance	taub	0.314	0.324	0.358	0.405	0.465	0.491	0.542	0.513	0.431	0.387	0.336	0.310	
	taud	2.507	2.477	2.409	2.267	2.151	2.117	1.999	2.070	2.286	2.381	2.481	2.554	
	Ebn at Noon	897	925	915	882	830	804	761	777	829	841	862	882	
	Edh at Noon	83	96	111	134	153	158	177	162	125	104	85	76	
All-Sky Solar Radiation	RadAvg	2.73	3.37	4.46	5.60	6.12	6.41	6.08	5.52	4.74	3.92	3.00	2.38	
	RadStd	0.15	0.36	0.34	0.40	0.49	0.44	0.37	0.26	0.40	0.47	0.24	0.21	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (69 neighbors)	+0.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+122

Nomenclature: See separate page